

◦ Aim:

- To determine the molecular weight of high polymer using viscosity measurement method.

◦ Apparatus:

- Ostwald's Viscometer, pipette, stopwatch, polymer solution.

◦ Theory:

- The average molecular weight of a polymer can be determined by a simple viscosity measurement. The viscosity of a high polymer solution depends on size and shape of molecule in solution. Hence, the measurement of viscosity is a useful method for study of polymer configurations. The method involves a preparation of a series of polymer solutions of different concentration in a suitable solvent and its viscosity is measured by using Ostwald's viscometer.
- If the absolute viscosity of the solution is ' η ' and that of the solvent is ' η_0 ', then relative viscosity is given by $\eta_r = \eta/\eta_0$. If flow time for solution and solvent are ' t ' and ' t_0 ', then relative viscosity, $\eta_r = t/t_0$ provided that the densities of solution and solvent are very close to each other, and hence it is necessary to work with very less density solutions. The specific viscosity is given as $\eta_{sp} = \eta_r - 1$ and the limiting / intrinsic viscosity $\eta = \eta_{sp}/c$. When η is plotted against a concentration c , a straight line should be obtained, which on extrapolation gives the limiting viscosity η . For linear polymers, the intrinsic viscosity is related to the molecular weight by Mark-Kuhn-Hawink Equation:

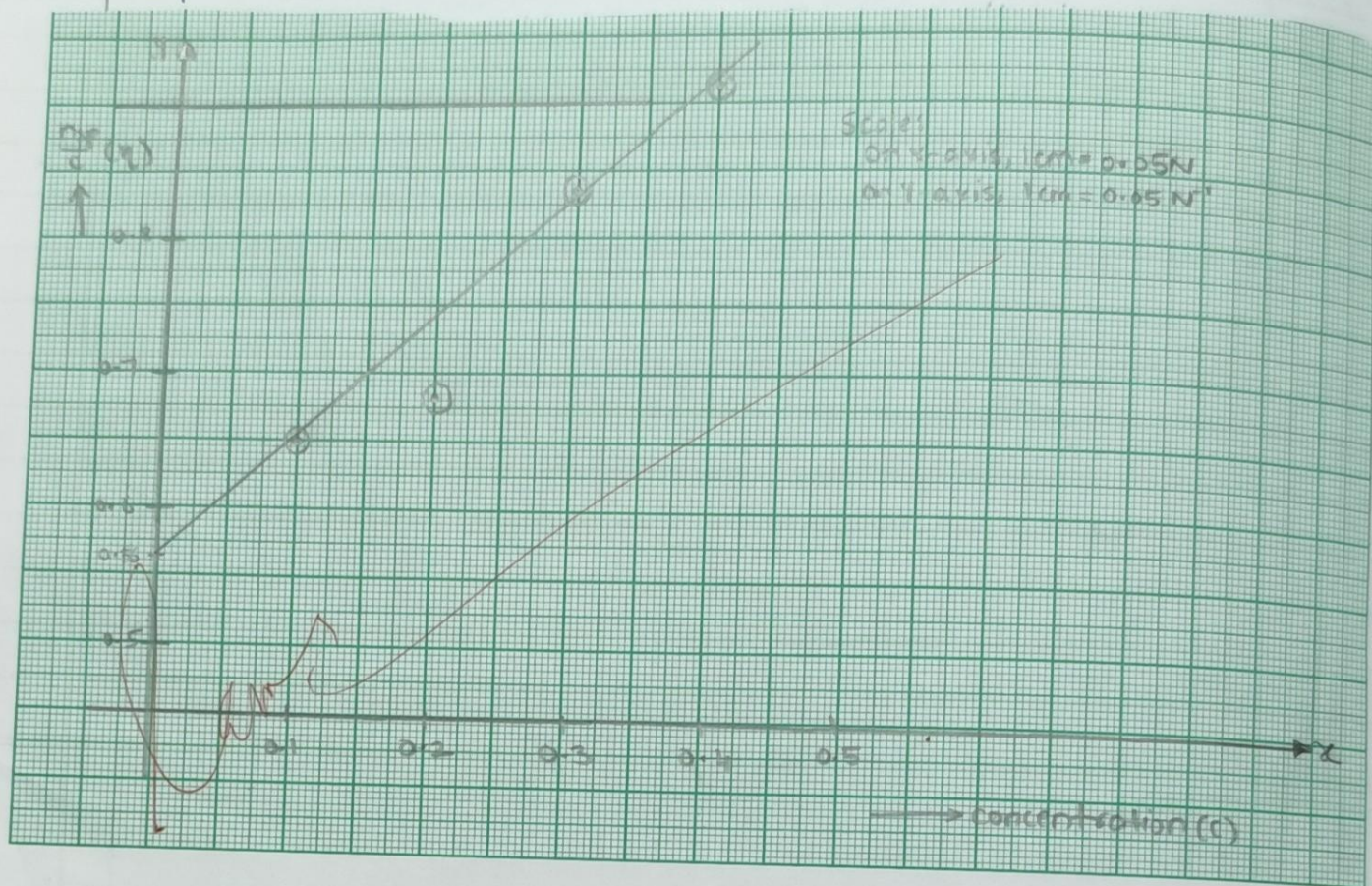
$$[\eta] = kM^a$$

M is the average molecular weight of the polymer.

° Observations:

Conc. (c)	Time of Flow (t)				Relative viscosity ($\eta_r = t/t_0$)	Specific viscosity ($\eta_{sp} = \eta_r - 1$)	Intrinsic viscosity ($\eta = \eta_{sp}/c$)
	t_1	t_2	t_3	t_{avg}			
Water	83	80	81	81.33s	—	—	—
0.1%	87	86	86	86.66s	1.065	0.065	0.65
0.2%	93	92	91	92s	1.131	0.131	0.66
0.3%	100	102	102	101.33s	1.245	0.245	0.82
0.4%	111	112	110	111s	1.364	0.364	0.912

Graph:



- The limiting viscosity is found out to be $\eta = 0.56$ since the solution had become too viscous to stir.

We know, $\eta = kM^a$

$$\therefore \log(\eta) = \log(k) + a \log(M)$$

$$\therefore \log(0.56) = \log(2 \times 10^{-4}) + 0.76 \log(M) \quad (k = 2 \times 10^{-4}, a = 0.76)$$

$$\therefore \log(M) = \frac{1}{0.76} (-0.251 + 3.698) = 4.535$$

$$\therefore M = 10^{4.535} = \underline{34318.343 \text{ gm/mol}}$$

M is the average molecular weight of the polymer.

K and a are constants for a given polymer-solvent system.
For polyvinyl alcohol-water system, $k = 2 \times 10^{-4}$, $a = 0.76$.

◦ Procedure:

- Varying concentration solutions are prepared (0.1%, 0.2%, 0.3%, 0.4%) by dissolving polymer in water.
- Viscosity is measured by using Ostwald's Viscometer for each solution.

◦ Calculations:

- We know, $N_1 V_1 = N_2 V_2$. Where $N_1 = 0.5$, $V_2 = 50 \text{ ml}$.

(i) When $N_2 = 0.1 \Rightarrow (V_1)_1 = 10 \text{ ml}$

(ii) When $N_2 = 0.2 \Rightarrow (V_1)_2 = 20 \text{ ml}$

(iii) When $N_2 = 0.3 \Rightarrow (V_1)_3 = 30 \text{ ml}$

(iv) When $N_2 = 0.4 \Rightarrow (V_1)_4 = 40 \text{ ml}$.

◦ Result:

- The molecular mass for given high polymer (PVA) is found to be 34318.343 gm/mol.